

All throughout this semester I have struggled with learning C++ at the same time I have been learning the course material. The background I had coming into the course was primarily in Python, so I had difficulty learning to properly apply principles of static typing, as well as navigating the intricacies of pointers, references, and addresses at the same time. I felt like I was starting to get the hang of the material when the assignments for OpenCL dropped on my head like a 100-lb steel plate. Because C code seems to be procedural instead of object-oriented (like C++ and python), my last thread to my previous coding experience had been severed. I am grateful that we were given several weeks to work on this assignment, because I don't know how I might have finished it otherwise. I tried to implement a simple program using vectors, buffers, and sub-buffers as we have been instructed in class, and the easiest way I thought to apply those principles was in the saxpy calculation as we saw in example scripts earlier in the course. I started with the 'simple.cpp' example script we were provided in Module 12, and I modified the kernel to apply the saxpy calculation using float4 vectors. Then I modified the program to create/allocate the necessary space to contain the new x,y, and a vectors. As I finished, I realized that almost the entire 'simple.cpp' program existed in the mainloop of the program. Because we are forbidden from using functions of more than 40 lines, my next task was gradually breaking the mainloop into a series of functions. My lack of experience with pointers/references really came to bite me in this process, but after a full day I finally completed the reduction and confirmed the functionality of the program.

```
• (base) desk1@pop-os:~/Documents/JHU/4.1/EN605.617/module12$ ./run.sh --platform 0 --useMap
In file included from /usr/include/CL/cl.h:20,
                 from info.hpp:23,
                 from assignment.cpp:12:
/usr/include/CL/cl_version.h:22:104: note: '#pragma message: cl_version.h: CL_TARGET_OPENCL_VERSION is not defined. Defaulting to 300 (OpenCL 3.0)'
 22 | #pragma message("cl_version.h: CL_TARGET_OPENCL_VERSION is not defined. Defaulting to 300 (OpenCL 3.0)")
    |
Number of platforms: 1
CL_PLATFORM_VENDOR: NVIDIA Corporation
CL_DEVICE_TYPE: CL_DEVICE_TYPE_GPU
(0.534639, 0.330193, 0.223558, 0.612253)
(0.100042, 0.203445, 0.82323, 0.28528)
(0.120521, 0.670136, 0.638715, 0.197932)
(0.943885, 0.602115, 0.634492, 0.199451)
(0.653463, 0.902053, 0.887333, 0.347533)
(0.814985, 0.464468, 0.765683, 0.027898)
(0.02821, 0.0178388, 0.737481, 0.0916929)
(0.177608, 0.680027, 0.334777, 0.189505)
(0.617652, 0.585655, 0.282449, 0.813604)
(0.697628, 0.0579379, 0.761754, 0.979821)
(0.967058, 0.850468, 0.492289, 0.496168)
(0.615727, 0.310915, 0.627176, 0.236567)
(0.21448, 0.29127, 0.340882, 0.0214005)
(0.139766, 0.0973159, 0.299287, 0.486746)
(0.953044, 0.210192, 0.877422, 0.959064)
(0.29347, 0.448009, 0.253598, 0.697595)
Program completed successfully
```